

Assignment 3: Interactive To-Do List App

Overview

Build a fully interactive to-do list application using vanilla JavaScript and the DOM. Tasks should persist between page reloads using localStorage. No frameworks or libraries are allowed for this assignment.

Learning Objectives

- Select, create, and modify DOM elements with JavaScript
- Handle user events (click, submit, keypress)
- Persist application state using the browser's localStorage API
- Use array methods (map/filter) to manage and render a list of items

Requirements

- Users can add a new task via a text input and button (or Enter key)
- Users can mark a task as complete/incomplete (visually distinct, e.g. strikethrough)
- Users can delete a task
- Tasks persist in localStorage and reload correctly when the page is refreshed
- The UI updates dynamically without a full page reload for any action
- Empty or whitespace-only tasks are rejected with user feedback
- The list is rendered from a JavaScript array/data structure, not hardcoded HTML

Technical Specifications

- No frameworks or UI libraries — vanilla JS, HTML, and CSS only
- JavaScript should be organized into named functions (e.g. addTask, renderTasks, deleteTask)
- Use at least one array method (map, filter, or reduce) in your rendering or update logic

Submission Guidelines

- Submit a GitHub repository link and zipped project folder
- Include at least two screenshots in your submission showing the page with no tasks and one with tasks
- Ensure the app runs by opening index.html directly (or note any required local server steps)

Grading Rubric

Criterion	Points	Description
Core functionality	30	Add, complete/toggle, and delete all work correctly
localStorage persistence	20	Tasks and their state survive a page reload
DOM manipulation quality	15	UI updates dynamically without full reloads; no leftover/duplicated elements
Input validation & edge cases	10	Empty tasks rejected; app doesn't break on rapid or repeated actions
Code organization & readability	15	Logical function breakdown, clear naming, minimal duplication
UI/UX polish	10	Clear visual states, reasonable styling, usable on first glance
Total	100	

Note: Submissions that do not run, fail to load, or throw uncaught errors on open may receive a substantial deduction across all criteria, since graders cannot evaluate functionality that isn't demonstrable.